NoteVision AI: Handwritten Note Digitization Using EMNIST-based CNN Classification

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Abstract - Handwritten text recognition is a valuable application of deep learning and computer vision, enabling computers to recognize and digitize handwritten text. This project describes NoteVision AI, a hybrid handwritten-note digitization system that utilizes an EMNIST-trained Convolutional Neural Network (CNN) alongside Microsoft TrOCR to identify handwritten text. EMNIST provides image classification at the character level using labelled 28 × 28 handwritten letters and digits while TrOCR is used to recognize complete handwritten text lines and enhance word-level readability. The EMNIST model learns visual patterns such as strokes, edges, shapes, and character structures while TrOCR employs an image Transformer encoder and text Transformer decoder. The system comprises image preprocessing, character and line segmentation, EMNIST-based classification, TrOCR-based line recognition, text reconstruction, confidence review, and PDF generation. This hybrid approach addresses the limitation of isolated-character classification when processing natural handwritten notes containing complete words and sentences. The recognized text can be displayed in an editable interface and exported as a downloadable PDF.

Keywords—handwritten text recognition, EMNIST, convolutional neural network, TrOCR, character segmentation, OCR, note digitization

I. Introduction

Handwritten notes are widely used in education, personal organization, and professional work but are challenging to search, edit, organize, and share when paper-based. Optical character recognition systems address this problem by transforming visual text into machine-readable content. Handwritten recognition is more challenging than recognition of printed text because character shapes can differ among writers, and are more susceptible to variation in writing speed, pen thickness, lighting, noise, skew, and overlapping strokes.

Recently, deep learning approaches have enabled Convolutional Neural Networks to achieve excellent results for image-based recognition tasks. The EMNIST dataset extends MNIST with handwritten letters and digits from the NIST Special Database 19 making it suitable for character-level classification [1]. However, natural handwritten notes are composed of complete words, variable spacing, and connected characters. To improve word-level recognition, NoteVision AI integrates Microsoft TrOCR, an encoder-decoder Transformer OCR model for text recognition. The microsoft/trocr-base-handwritten model is fine-tuned on the IAM handwriting dataset and is designed for single text-line images [6]. In the proposed system, EMNIST remains a part of the core visual classification and segmentation pipeline while TrOCR reads segmented text lines as sequences. The application processes an uploaded handwritten note image and generates editable digital text that can be exported as a PDF.

The main objective of this work is to design an EMNIST-based handwritten note digitization pipeline that incorporates image preprocessing, character segmentation, CNN-based classification, text reconstruction, confidence analysis, and PDF generation within a single user-friendly application.

II. Related Work

Cohen et al. introduced EMNIST as an extension of MNIST for handwritten letters and digits. The dataset preserves the 28 × 28 image structure familiar from MNIST while extending the classification task to handmade alphanumeric characters [1]. EMNIST provides multiple dataset splits including ByClass, ByMerge, Balanced, Letters, and Digits, allowing researchers to choose an appropriate classification problem [2].

The usage of CNN-based handwritten character recognition has been well studied as convolutional layers can learn visual features directly from image pixels. Chooi and Ghafar demonstrated the application of CNN methods for handwritten character recognition using EMNIST while highlighting the usefulness of TensorFlow-based deep-learning workflows for this task [3].

Character recognition alone is inadequate for complete handwritten documents, a document-level system is required to determine where characters occur and reconstruct them in the correct reading order. Jo et al. described the importance of handwritten text segmentation and showed that segmentation remains a major challenge in document processing [4]. This is particularly relevant to NoteVision AI because connected handwriting and inconsistent spacing can cause multiple characters to be merged or incorrectly separated.

These studies motivate a design for NoteVision AI as a hybrid pipeline with classical computer-vision preprocessing, the trained EMNIST CNN for character-level visual classification, and Microsoft TrOCR for sequence-level handwritten line recognition. Microsoft TrOCR combines an image Transformer encoder and text Transformer decoder and generates text tokens autoregressively [6], [7]. This combination allows EMNIST to remain an explicit part of the project while using a more powerful word- and line-level OCR component for natural handwritten notes. The output is then reconstructed into editable text for user review.

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III. Proposed System

A. System Architecture

The NoteVision AI architecture comprises seven major stages: image upload, preprocessing, character and line segmentation, EMNIST CNN classification, TrOCR line recognition, text reconstruction and confidence review, and PDF generation. The system is implemented using Python. Streamlit provides the web interface, OpenCV performs image processing, TensorFlow loads and performs the trained EMNIST CNN, Hugging Face Transformers library loads Microsoft TrOCR, and a PDF generation module creates the final downloadable document.

B. Image Preprocessing

The uploaded handwritten image is transformed into a suitable form for analysis. Typical preprocessing operations include grayscale conversion, noise reduction, contrast enhancement, thresholding, and inversion if needed. The purpose of preprocessing is to maximize the contrast between handwritten strokes and the background. Since EMNIST samples are grayscale character images, the segmented input must ultimately be transformed into a representation compatible with the CNN.

C. Character Segmentation

OpenCV-based image analysis is used to detect handwritten regions and generate bounding boxes around potential characters. Each detected region is cropped, normalized, and resized to 28 × 28 pixels. The bounding boxes are sorted in accordance with their vertical and horizontal position so that characters can be reconstructed in approximate reading order. Line grouping is important in this step to avoid simply sorting every bounding box from left to right, since that would mix characters from different lines.

D. EMNIST-based CNN Classification

The core learning component is a CNN trained on EMNIST handwritten characters. During inference, each normalized character image is passed to the model, which produces probabilities for the available classes. The class with the maximum probability is the predicted character while the top alternatives are kept for confidence analysis. This approach meets the image-classification requirement of the system since recognition is performed by classifying each handwritten character image.

E. Text Reconstruction and Confidence Review

Predicted characters are combined according to their ordered bounding boxes to form reconstructed text. The application displays the recognized output in an editable text area, where users can correct errors before exporting. Predictions are also grouped according to their confidence levels. Predictions with low confidence can be highlighted with alternate character predictions. This human-in-the-loop step enhances practical usability since character-level classifiers may confuse visually similar handwritten symbols.

F. PDF Generation

After reviewing the recognized text, the user can generate a digital PDF. The generated document contains the corrected text rather than forcing the user to accept raw model output. This design makes NoteVision AI useful even when the recognition stage is imperfect.

G. Hybrid EMNIST and TrOCR Recognition

EMNIST and TrOCR contribute together to the system's performance. The EMNIST CNN operates on segmented character regions to provide character-level predictions, class probabilities, and confidence information. The detected line regions are then passed to Microsoft TrOCR for recognition of the complete handwritten line as a text sequence. This reduces the reliance on independent character predictions for word formation and helps maintain natural spelling and word context. The final application retains the EMNIST outputs available for detection, classification, and confidence analysis while employing TrOCR as the main line-level text reader.

IV. Methodology

A. Dataset

The EMNIST dataset is used as the foundation for training the character classifier. EMNIST contains handwritten characters drawn from NIST data as 28 × 28 grayscale images [1], [2]. Depending on the selected split, the dataset can represent digits, letters or a combination of uppercase letters, lowercase letters and digits.

B. Training Pipeline

The training pipeline loads the selected EMNIST samples, normalizes pixel values, applies the required orientation correction, and prepares the labels for multiclass classification. Orientation handling is particularly important since standard EMNIST data may appear rotated and transposed in its stored representation [2]. The CNN learns spatial features using convolutional layers and produces a probability distribution over the character classes.

C. Inference Pipeline

For an uploaded note, first preprocessing is applied to the complete page, followed by extraction of candidate character regions and line groupings. Each character candidate can be classified by the EMNIST CNN for visual classification and confidence analysis. The corresponding line crops are then passed to Microsoft TrOCR for sequence-level handwriting recognition. The TrOCR outputs are combined in top-to-bottom line order to form the final text. This hybrid inference stage reduces the effects of individual character errors while retaining EMNIST as a core image-classification component. The final output is presented for user editing before creating a PDF.

D. TrOCR-based Line Recognition

The Microsoft TrOCR handwritten model is used as a complementary OCR component. TrOCR utilizes a Transformer-based vision encoder and text decoder to map an input text-line image to a generated text sequence [7]. The project uses the microsoft/trocr-base-handwritten checkpoint, which is fine-tuned on IAM handwriting data, and is intended for single-line OCR [6]. Instead of predicting every letter independently, TrOCR processes a complete segmented line, allowing the model to use contextual information when generating words. This is particularly valuable for natural handwritten notes where isolated EMNIST character classification can lead to errors in word boundaries or visually similar characters.

V. Implementation

The NoteVision AI prototype is organized into modular components. The preprocessing module reads the uploaded

images and prepares them for segmentation. The segmentation module detects candidate characters, records bounding boxes, and groups regions into lines. The recognition module loads the EMNIST-trained TensorFlow model and maps predicted class indices to characters. A separate TrOCR recognition module loads microsoft/trocr-base-handwritten through the Transformers library and recognizes complete text lines. The main Streamlit application manages the workflow and displays the original image, processed image, detected regions, recognized text, confidence information, and PDF download option.

This modular structure enhances maintainability and allows individual stages to be improved independently. For example, segmentation can be enhanced without replacing the trained CNN, and the recognition model can later be retrained using additional handwriting samples without redesigning the interface.

VI. Challenges and Discussion

The main limitation encountered in practical testing is the domain gap between EMNIST and real handwritten notes. EMNIST primarily supports isolated character classification, whereas natural notes contain connected letters, variable spacing, complete words, multiple lines, diagrams, and uneven illumination. As a result, a character classifier may correctly recognize isolated segments but still produce incorrect words due to segmentation or reading order errors. Microsoft TrOCR improves the word- and line-level recognition stage as it generates text sequences from complete handwritten lines but remains dependent on the quality of line segmentation, image preprocessing, handwriting style, and input quality.

Visually similar characters also cause ambiguities. Depending on handwriting style, characters such as O and 0, I and l, or uppercase and lowercase forms may be confused. Incorrect segmentation can also cause one character to be split into multiple regions or multiple characters to be treated as one region.

The proposed application addresses these issues through image preprocessing, ordered line-aware bounding boxes, EMNIST classification and confidence analysis, TrOCR line recognition, and user editing. EMNIST is therefore retained as an important image-classification component rather than being removed from the pipeline, while TrOCR provides an additional sequence-level recognition stage for improved readability. The two models have different responsibilities and are employed together to bridge the gap between isolated-character datasets and complete handwritten notes.

VII. Future Enhancements

Future work can enhance recognition by fine-tuning TrOCR on a larger collection of real academic handwritten notes, training EMNIST with additional handwriting samples, applying data augmentation, improving line and word segmentation, and using language-based post-processing to correct improbable character sequences. A domain-specific language model can be utilized after TrOCR recognition while preserving EMNIST as the core visual classification model. Support for multi-page documents, automatic paragraph detection, layout preservation, and joint confidence estimation between EMNIST and TrOCR could further improve the application.

VIII. Conclusion

This paper presented NoteVision AI, a hybrid handwritten note digitization system that combines EMNIST-based CNN image classification with Microsoft TrOCR line-level handwriting recognition. The system comprises image preprocessing, OpenCV-based segmentation, character classification, Transformer-based sequence recognition, ordered text reconstruction, confidence review, and PDF generation. EMNIST provides the required character-level image-classification component while TrOCR aids recognizing complete handwritten words and lines more naturally than independent character predictions. The project demonstrates that combining complementary recognition models can provide a valuable bridge between character-level handwritten datasets and real handwritten documents. The final workflow allows users to review and edit the recognized text prior to generating downloadable digital notes.

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